

# Task – COVID Data Analysis using NumPy, Pandas & Visualization

## Scenario

You are working as a Data Analyst in a healthcare organization.

You have been given COVID-19 data in CSV format.

Perform data cleaning, analysis, and visualization using NumPy, Pandas, Matplotlib, and Seaborn.

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## Topics to Cover

### NumPy

- Single Dimensional Array
  - Multi Dimensional Array
  - ndarray
  - Mathematical Operations
  - Vectorized Operations
  - Broadcasting
  - Slicing & Indexing
  - Reshaping Arrays
  - Splitting Arrays
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### Pandas

- Series
- DataFrame
- Reading CSV file
- Slicing & Iterating
- Statistical Operations
- Handling Missing Values
- Handling Duplicate Values

- GroupBy Operations
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## Visualization

Use:

- Matplotlib
- Seaborn

Create:

- Line Graph
  - Bar Chart
  - Histogram
  - Pie Chart
  - Scatter Plot
  - Heatmap
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## Tasks

### Task 1 – NumPy Operations

- Create NumPy arrays using COVID confirmed cases data.
  - Perform:
    - Mean
    - Sum
    - Max
    - Min
    - Reshaping
    - Slicing
    - Broadcasting
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### Task 2 – Read CSV using Pandas

- Read the COVID dataset using Pandas.
- Display:

- First 5 rows
  - Last 5 rows
  - Data types
  - Null values
  - Duplicate values
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## Task 3 – Data Cleaning

- Handle missing values.
  - Remove duplicate rows.
  - Replace invalid values like:
    - inf
    - null
    - 0 where necessary
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## Task 4 – Exploratory Data Analysis (EDA)

Perform analysis on:

- Total confirmed cases
- Total deaths
- Total recovered cases
- WHO region wise analysis
- Highest active cases
- Top 10 affected regions

Use:

- `groupby()`
  - sorting
  - filtering
  - aggregation functions
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## Task 5 – Visualization

Create the following charts:

- Line graph for confirmed cases

- Bar chart for WHO region wise deaths
  - Histogram for active cases
  - Pie chart for recovered cases by region
  - Scatter plot between confirmed and deaths
  - Heatmap for correlation analysis
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## Expected Output

- Cleaned COVID dataset
- Statistical summary
- Region-wise analysis
- Different visualizations
- Insights from data